Identifying Patterns in Global E-Commerce Sales

Chosen problem statement: Option C

Group members and student IDs: [TO BE COMPLETED]

Course title: [TO BE COMPLETED]

Submission date: [TO BE COMPLETED]

Transaction patterns, customer segmentation and business actions

Ethics and Integrity Declaration: signed group form required by the brief; completion not verified.

Executive Summary

We analysed the supplied retail transactions to connect purchasing patterns with customer actions. Chronological cancellation matching leaves GBP 16,338,578.60 in identified-customer net merchandise value. Missing IDs exclude GBP 2,575,278.90 of otherwise valid purchase value, so these results describe observable customers rather than the whole business. RFM with 2 clusters achieves silhouette 0.438, supporting a coarse behavioural split rather than perfectly separated customer types. The separate high-value group contributes 27.54% of net value. We recommend differentiated retention tests and individual major-account review, supported by better customer identification and return linkage. Revenue concentration does not establish profitability or campaign effectiveness.

Introduction

Option C asks how a UK online retailer can use transaction records to understand products, seasonality, countries and customers. The business question is practical: which purchasing patterns warrant a different merchandising or customer-contact decision? We use segmentation as an analytical model that organises observed behaviour. There is no supervised outcome, and forecasting is outside this project's scope.

The supplied Online Retail II workbook spans 2009-12-01 07:45:00 to 2011-12-09 12:50:00. It is the sole empirical data source; the supplied assignment PDF defines the task. The UCI source is identified in the brief (Online Retail II, 2019). We combine descriptive analysis with a reproducible scikit-learn pipeline, preserving distinctions between gross purchases, retrospective net sales and invoice cancellation rates. The output is a basis for controlled business tests, not evidence that customers will respond to a particular intervention.

The proposed business use is to allocate attention more deliberately. A merchandising team might prioritise availability for products with repeated demand, while a customer team could vary contact according to observed activity. These decisions require different evidence. We therefore keep product-level demand, country-level concentration and customer-level segmentation distinct, and link recommendations to the specific summaries that support them. Customer spending is an outcome to describe, not an explanation of motivation.

Data Description and Preprocessing

The workbook contains 1,067,371 rows and 8 columns across the source sheets. Fields identify invoices, products, descriptions, quantities, dates, prices, customers and countries. The sheet date ranges overlap from 2010-12-01 08:26:00 to 2010-12-09 20:01:00. Exact deduplication removes 34,335 rows. Schema, null counts and date ranges are saved by sheet in the results file. We use pandas and NumPy for data handling (The pandas development team, n.d.; NumPy developers, n.d.). The complete sequential cleaning audit follows this narrative.

Before extending exclusions, the original code list passes every supplied regression check. We then remove 237 additional fee, sample and voucher rows. An evidence table preserves irregular codes and example descriptions: syntax alone never proves a code is invalid. Purchases require positive quantity and price. Missing IDs are not imputed: 226,761 valid purchase rows, 13.11% of gross valid merchandise value, are excluded from customer linkage. This coverage loss may systematically distort customer and country profiles.

Cancellations reverse only strictly earlier purchases with the same customer and stock code, taking the most recent eligible units first. Remaining units keep the original purchase price. This is an explicit allocation assumption because invoice-level return links are unavailable. Of 17,914 cancellation rows, 15,872 are matched, 127 partly matched and 1,915 unmatched. We reverse 450,047 units and GBP 730,004.12; 17,692 units remain unmatched. Netting drops 6,026 fully reversed lines. Unmatched returns are reported without deducting them from unrelated purchases. Extreme purchases specified in the request are fully reversed by automated tests.

Complete cleaning audit

Rows before	Removed	Rows after	Reason / treatment
1,067,371	34,335	1,033,036	Remove exact duplicate rows across both sheets, including their overlapping dates; no fuzzy deduplication.
1,033,036	5,581	1,027,455	Remove only original non-product code list, after uppercase and whitespace normalisation; fixed-list regression passed.
1,027,455	237	1,027,218	Extend exclusions only with documented fee, sample and gift-voucher codes; evidence is in non_product_evidence.csv.
1,027,218	17,914	1,009,304	Separate all cancellation-invoice rows into a retained cancellation table; these are not booked as sales.
1,009,304	5,964	1,003,340	Remove non-cancellation rows with nonpositive quantity or price; no price imputation.
1,003,340	226,761	776,579	Exclude missing-ID positive purchase rows from linked customer analysis; their gross merchandise value and coverage loss are reported separately.
776,579	6,026	770,553	Drop fully reversed purchase lines after chronological customer/product LIFO matching; partly reversed lines retain remaining units at original purchase prices. Unmatched cancellation units remain outside net sales.

Exploratory Data Analysis

Figure 1 shows the actual peak, November 2011, at GBP 1,119,164.27. Complete-month fourth-quarter mean revenue is 1.53 times the other-month mean. The incomplete final month is shaded and excluded from this comparison. Matching returns to original sales means the curve describes retained purchase cohorts, not cash movements on return dates. The short window supports a descriptive seasonal pattern, not a general seasonal law. Figures use Matplotlib and seaborn (Matplotlib development team, n.d.; seaborn developers, n.d.).

Figure 2 separates value, volume and reach. REGENCY CAKESTAND 3 TIER leads revenue at GBP 261,232.75; its product value per invoice is GBP 80.65, with 575 repeat customers. WORLD WAR 2 GLIDERS ASSTD DESIGNS leads volume at 103,697 units. WHITE HANGING HEART T-LIGHT HOLDER reaches 1,484 customers, with 55.5% returning for the product. These different leaders justify distinguishing stock throughput, transaction value and repeat demand when setting priorities.

Figure 3 excludes the United Kingdom by name. EIRE leads export revenue with GBP 567,785.75 from only 3 customers. Netherlands averages GBP 2,538.50 per order and 1,774.2 units per order, versus UK values of GBP 411.79 and 247.3. Larger observed baskets accompany high order value, while tiny customer bases limit generalisation. We cannot infer wholesale status. Figure 4 reports complete-month mean cancellation invoice rates of 16.21% in the fourth quarter and 17.03% otherwise; there is no descriptive fourth-quarter elevation.

Modelling Approach

Clustering inputs, in fitted order: Recency, Frequency, Monetary. All customer features are Recency, Frequency, Monetary, AOV, Variety, CancellationRate. Recency is whole days since the latest retained purchase, measured at 2011-12-10 12:50:00; Frequency counts retained invoices; Monetary sums retained merchandise value. AOV equals Monetary divided by Frequency, Variety counts products, and CancellationRate divides distinct cancellation invoices by cancellation plus valid gross purchase invoices for identified customers. Fully reversed-only customers do not enter the positive-net customer population.

AOV is omitted from clustering because it algebraically repeats information already present. CancellationRate is zero for 58.8% of customers; its skew changes from 1.32 to 1.14 under log transformation. It remains a profiling measure. Figure 5 and the correlation heatmap show transformation behaviour and redundancy. We apply the conventional upper Tukey fence to log net Monetary, recalculated after netting: GBP 36,098.33. The 40 customers above it remain visible in Outlier-Wholesale; that requested label is not verified buyer type.

We compare RFM with an extended set adding Variety across candidate cluster counts, recording inertia, full-sample silhouette, Davies-Bouldin and mean pairwise adjusted Rand agreement across 10 seeds. Highest silhouette determines the choice; the full table and Figure 6 expose the alternatives. The chosen rfm model has silhouette 0.438, Davies-Bouldin 0.873 and seed agreement 0.999. More clusters may support more actions, but these results do not justify trading away separation without an operational test. The pipeline applies log transformation, standardisation and K-Means together (Pedregosa et al., 2011). Cluster IDs are reordered by descending mean Monetary, and saved-model reload predictions are checked.

The evaluation measures answer different questions. Inertia records within-group squared distance and generally decreases as groups are added. Silhouette compares cohesion with distance to neighbouring groups. Davies-Bouldin summarises similarity between groups, where lower values indicate more separation. Adjusted Rand agreement compares assignments across starting seeds after allowing for relabelling. We examine these together because a visually attractive projection or falling inertia alone cannot establish a useful partition. There is no labelled ground truth against which to report classification accuracy.

Results and Discussion

Figure 7 projects the transformed customers into principal components, retaining 94.8% of variance. Projection overlap reinforces the need for measured language. Figure 8 standardises raw segment means against the all-customer population, including outliers. The profile tables report both means and medians to expose skew. Names use segment medians relative to all-customer terciles; automated checks reject names that contradict recency, frequency or monetary tags.

Loyal champions contains 2,347 customers, 40.20% of customers and 60.35% of net revenue (GBP 9,860,691.30). Median recency is 25 days and median frequency is 8 invoices. Offer early access to relevant products and test a replenishment reminder against a holdout group.

Dormant occasional buyers contains 3,452 customers, 59.12% of customers and 12.10% of net revenue (GBP 1,977,598.21). Median recency is 297 days and median frequency is 2 invoices. Test a low-cost reminder using their last product category before committing a wider recovery budget.

Outlier-Wholesale contains 40 customers, 0.69% of customers and 27.54% of net revenue (GBP 4,500,289.09). Median recency is 4 days and median frequency is 65 invoices. Review major accounts individually, confirm buyer needs and set stock reservations against their observed repeat orders.

The saved pipeline classifies a synthetic median-based customer with recency 96 days, frequency 3, Monetary GBP 837 and internally consistent AOV GBP 279 as Dormant occasional buyers. This checks artifact use, not predictive accuracy. A deployment wrapper must apply the separate outlier fence before calling the fitted ordinary-customer model.

Segment sizes and revenue

Segment	Customers	Customer share	Net value (GBP)	Revenue share
Loyal champions	2,347	40.20%	9,860,691.30	60.35%
Dormant occasional buyers	3,452	59.12%	1,977,598.21	12.10%
Outlier-Wholesale	40	0.69%	4,500,289.09	27.54%

Conclusion and Recommendations

The main recommendation is to preserve valuable active relationships, review exceptional accounts individually, and test economical reminders for the dormant group. Group size alone should not dictate budget: revenue concentration, contact costs, margins and incremental response all matter. Product planning should likewise distinguish high-unit products from products with higher transaction value and repeated customer demand. Export decisions should examine account concentration before extrapolating a country average to a new market.

What the data cannot tell us is substantial. Missing IDs remove 226,761 purchase rows, 22.60% of valid purchase rows and GBP 2,575,278.90, or 13.11% of gross valid purchase value. Unmatched cancelled units account for 3.78% of cancelled units. There are no browsing events, marketing spend, margins or return reasons. Cancellation invoices are observable, but a complete physical-return rate is not. We cannot distinguish customers who left from customers whose normal buying cycles are long, or identify the reason for inactivity.

The evidence comes from a single retailer and a historical observation window with an incomplete final month. RFM is backward-looking, and different customers have different opportunities to accumulate orders. Euclidean K-Means favours compact groups after transformation; changing the observation window, fence or feature weighting could change membership. Strong seed agreement is not temporal stability or evidence of campaign uplift. Correlation is not causation, and a high-value segment need not be profitable.

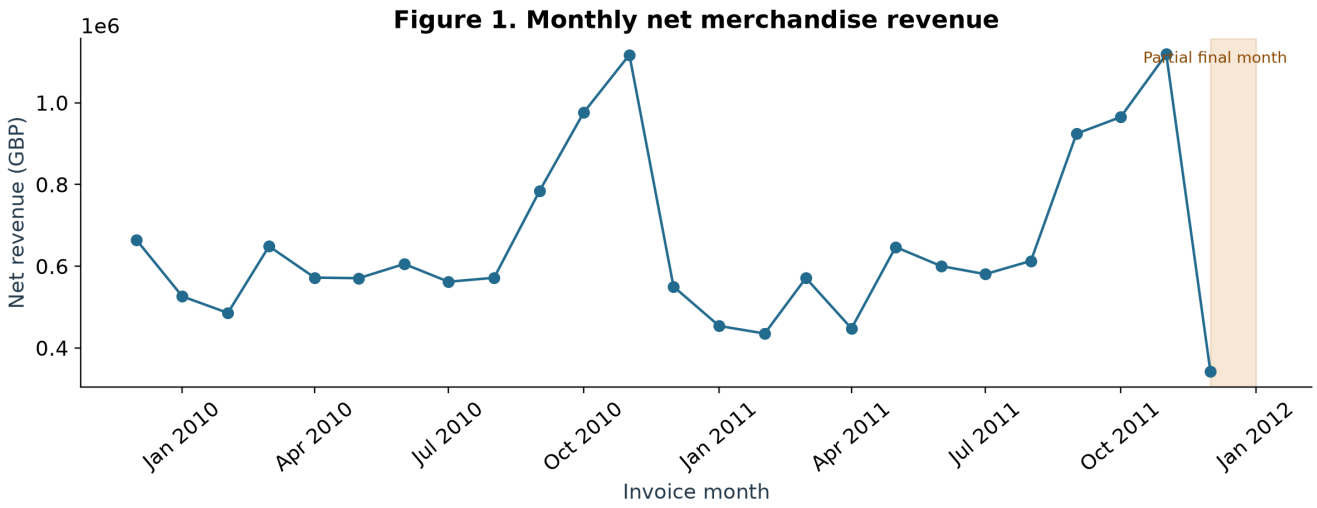
Before operational use, improve customer-ID capture and explicit return-to-sale links, verify the high-value accounts, and run controlled action tests with untreated comparisons. Measure incremental net purchasing after returns and costs, then revisit segment usefulness. AI assisted code generation, debugging, report drafting and verification; human group members must understand and review the work. The source citation remains provisional pending verification on the UCI page. No external retail facts were added, and no new customer response or profit estimates were invented.

A practical follow-up would track whether the proposed contact changes retained purchasing without simply subsidising orders that would have happened anyway. Contact frequency, stock availability and opt-out behaviour should also be considered before scaling. Those variables are absent here, so the report proposes a test design rather than estimating an effect. Rebuild the features consistently at later snapshots and compare segment movement before treating the current groups as permanent customer identities.

Raw customer profiles

Segment / units	Feature	Mean	Median
Loyal champions	Recency (days)	53.68	25.00
Loyal champions	Frequency (count)	10.86	8.00
Loyal champions	Monetary (GBP)	4,201.40	2,610.28
Loyal champions	AOV (GBP)	411.62	326.38
Loyal champions	Variety (count)	143.84	111.00
Loyal champions	CancellationRate (invoice ratio)	13.99%	12.50%
Dormant occasional buyers	Recency (days)	302.06	297.00
Dormant occasional buyers	Frequency (count)	2.03	2.00
Dormant occasional buyers	Monetary (GBP)	572.88	393.82
Dormant occasional buyers	AOV (GBP)	307.16	228.10
Dormant occasional buyers	Variety (count)	34.58	25.00
Dormant occasional buyers	CancellationRate (invoice ratio)	9.01%	0.00%
Outlier-Wholesale	Recency (days)	32.23	4.00
Outlier-Wholesale	Frequency (count)	95.72	65.00
Outlier-Wholesale	Monetary (GBP)	112,507.23	61,911.35
Outlier-Wholesale	AOV (GBP)	1,585.23	977.56
Outlier-Wholesale	Variety (count)	523.98	249.00
Outlier-Wholesale	CancellationRate (invoice ratio)	14.64%	14.17%

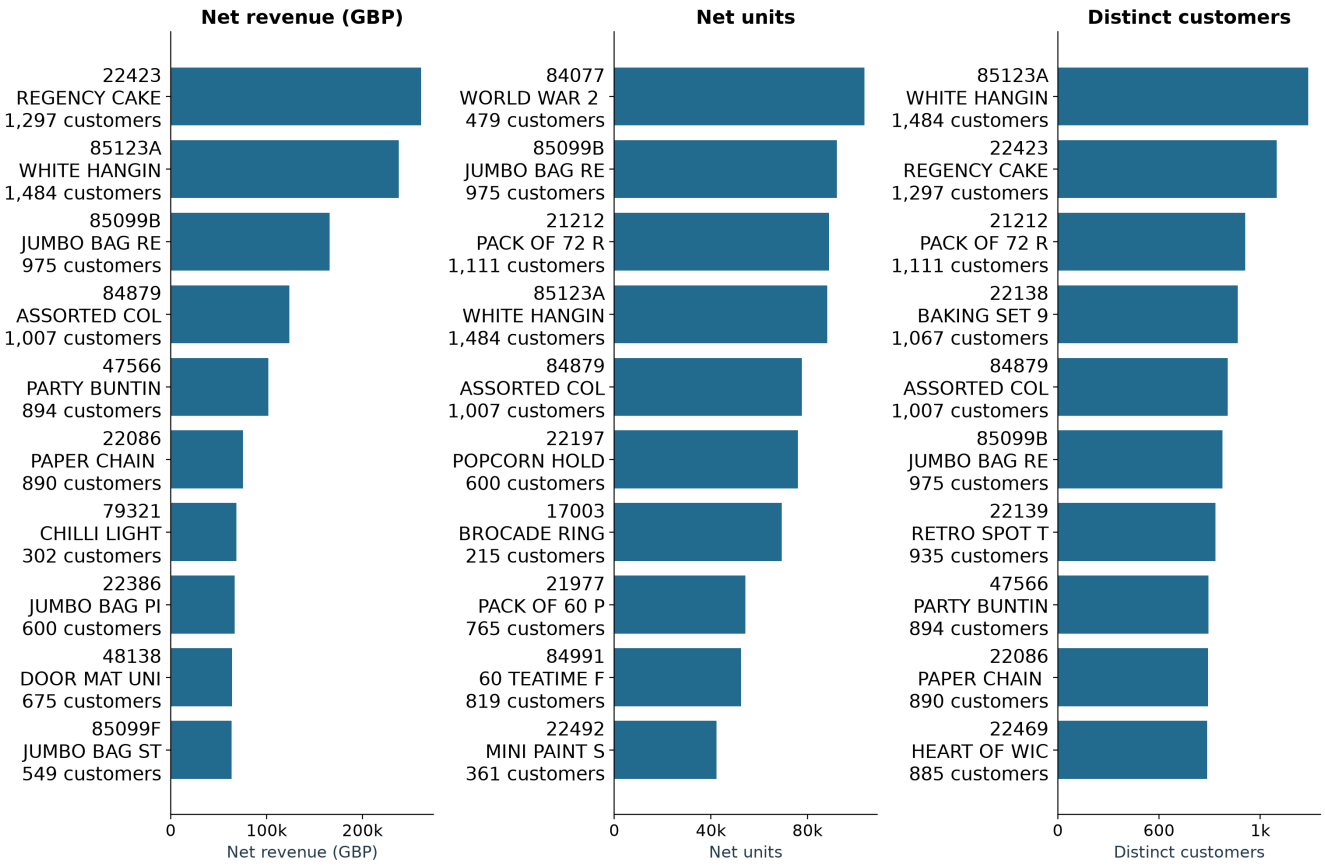
Figure 1. Monthly net merchandise revenue



Peak months: November 2011 (GBP 1,119,164), November 2010 (GBP 1,117,670), October 2010 (GBP 976,736). Complete-month Q4/rest mean revenue ratio: 1.53. December 2011 ends 09 December 2011. Reversals are attributed to the original sale month.

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Figure 2. Products ranked separately by each measure

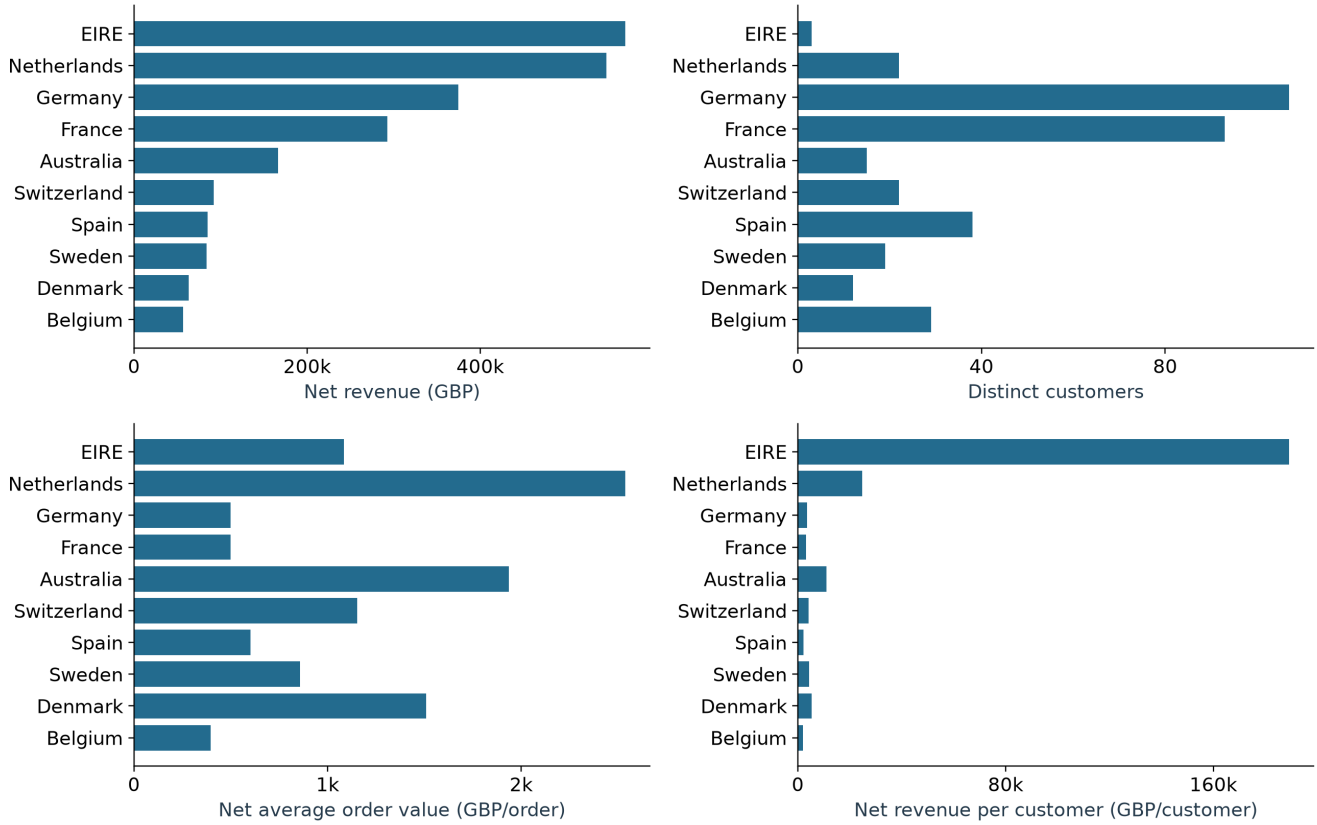


revenue leader: REGENCY CAKESTAND 3 TIER (22423), 261,233 GBP; units leader: WORLD WAR 2 GLIDERS ASSTD DESIGNS (84077), 103,697 units; customers leader: WHITE HANGING HEART T-LIGHT HOLDER (85123A), 1,484 customers. The leaders differ. Each row shows its customer count; full names are in product_summary.csv.

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Figure 3. Highest-revenue export countries (United Kingdom excluded)

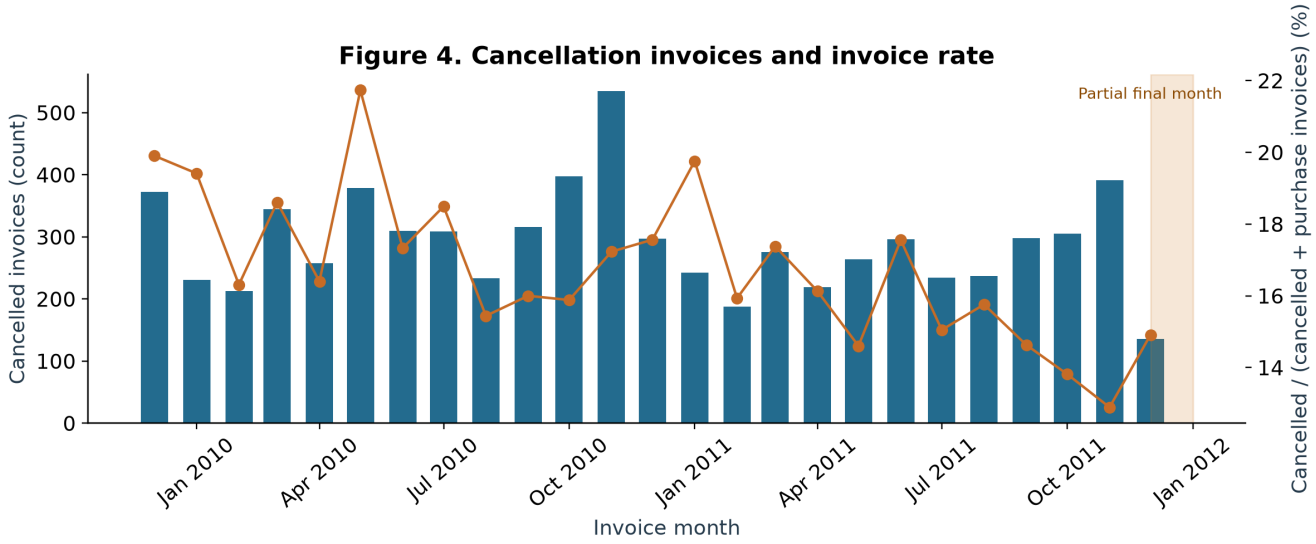
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EIRE leads exports: GBP 567,786, 3 customers. Netherlands has the largest AOV in this group: GBP 2,538.50, 1,774.2 units/order, 22 customers; UK: GBP 411.79, 247.3 units/order. Basket size and customer-base size are descriptive, not proof of buyer type.

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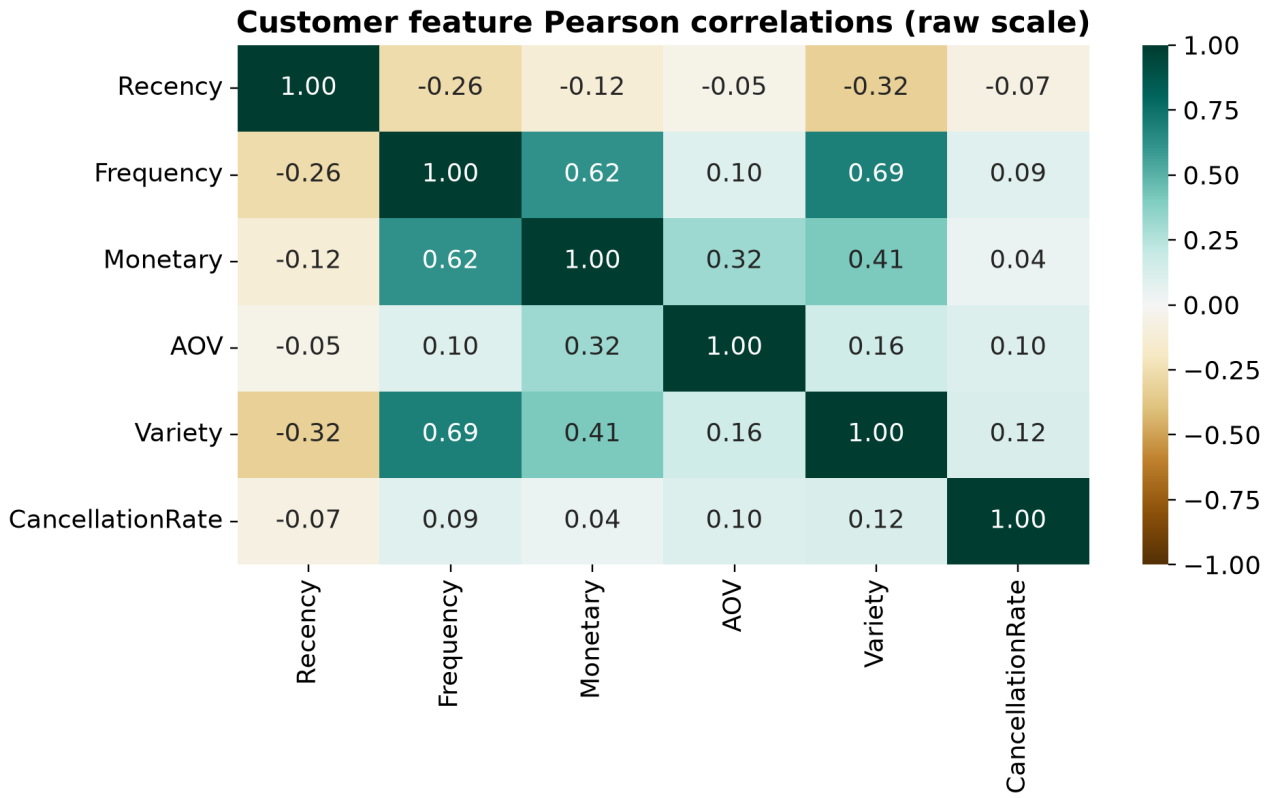
Figure 4. Cancellation invoices and invoice rate



Complete-month mean invoice cancellation rate: Q4 16.21%; non-Q4 17.03%. No descriptive Q4 elevation is observed. Both invoice counts use identified customers and merchandise codes; gross purchase invoices precede netting. Final month is partial.

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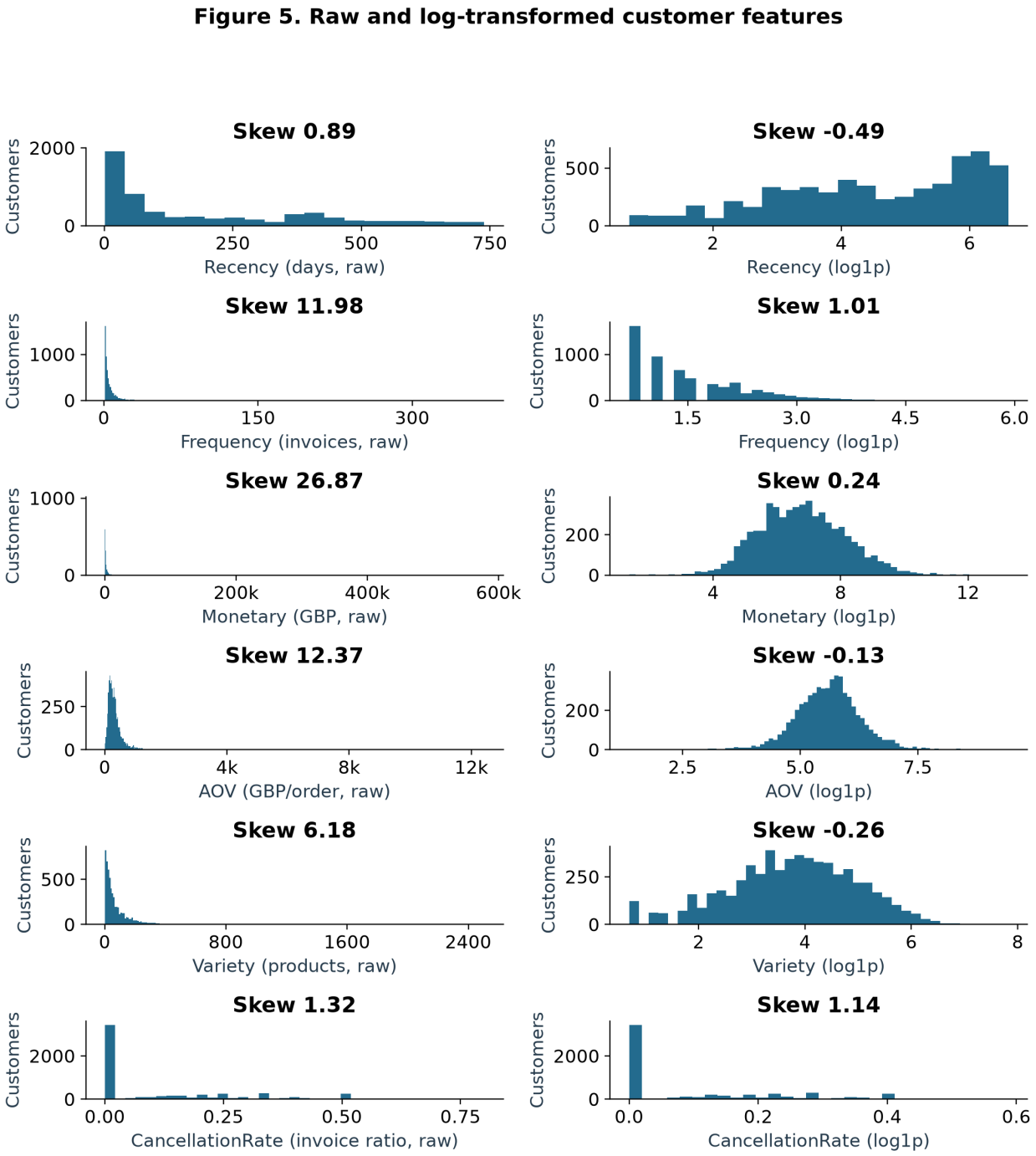
Customer feature Pearson correlations (raw scale)



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Figure 5. Raw and log-transformed customer features

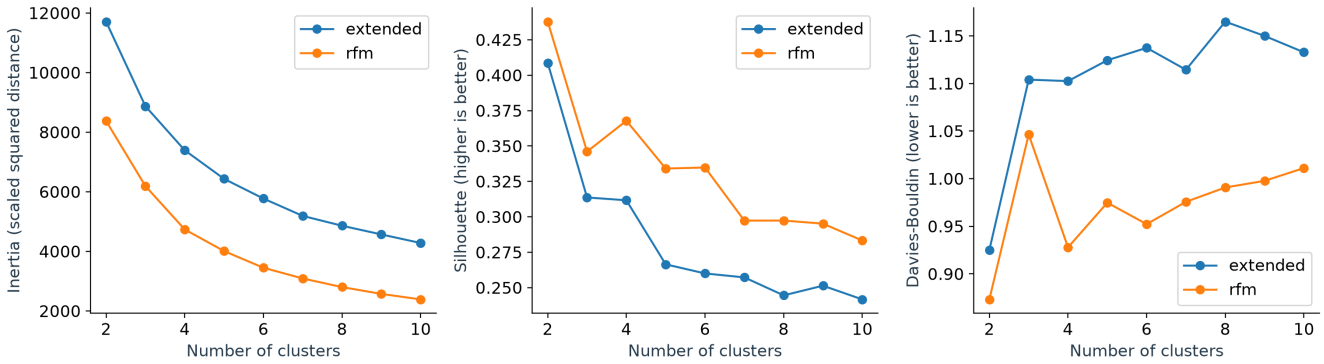


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Figure 6. Feature-set and cluster-count comparison

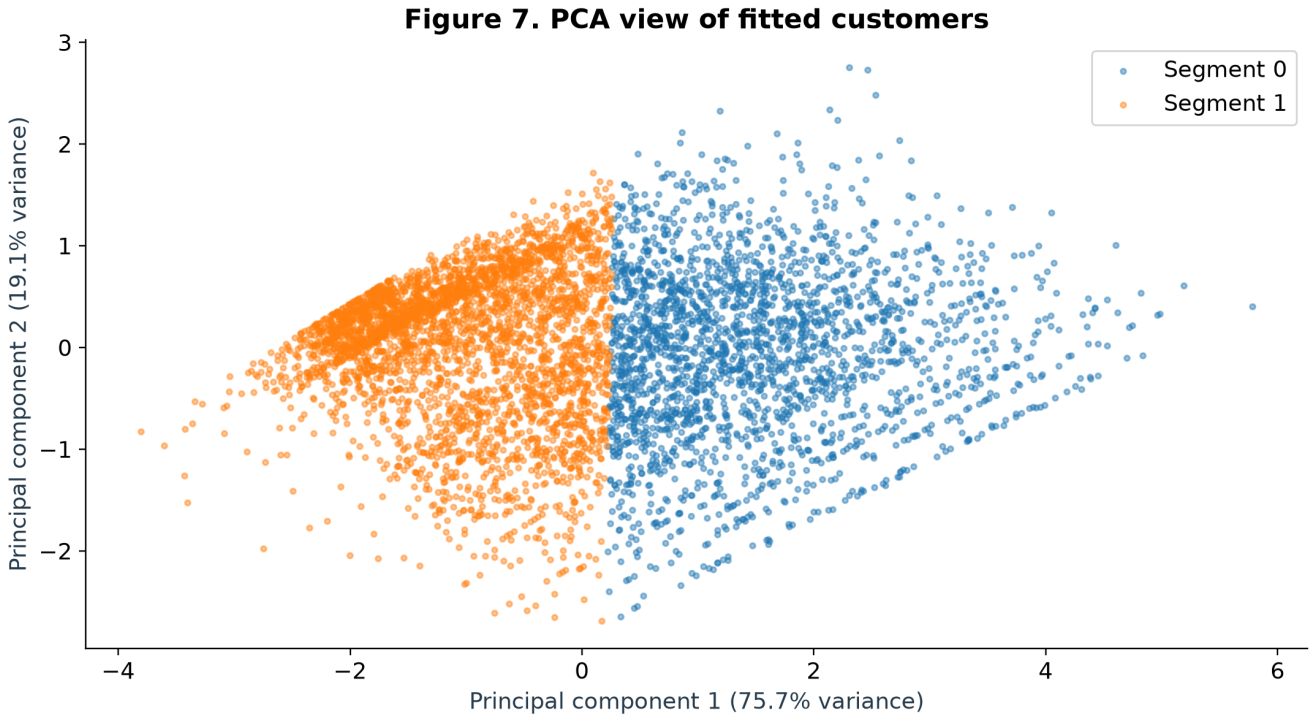
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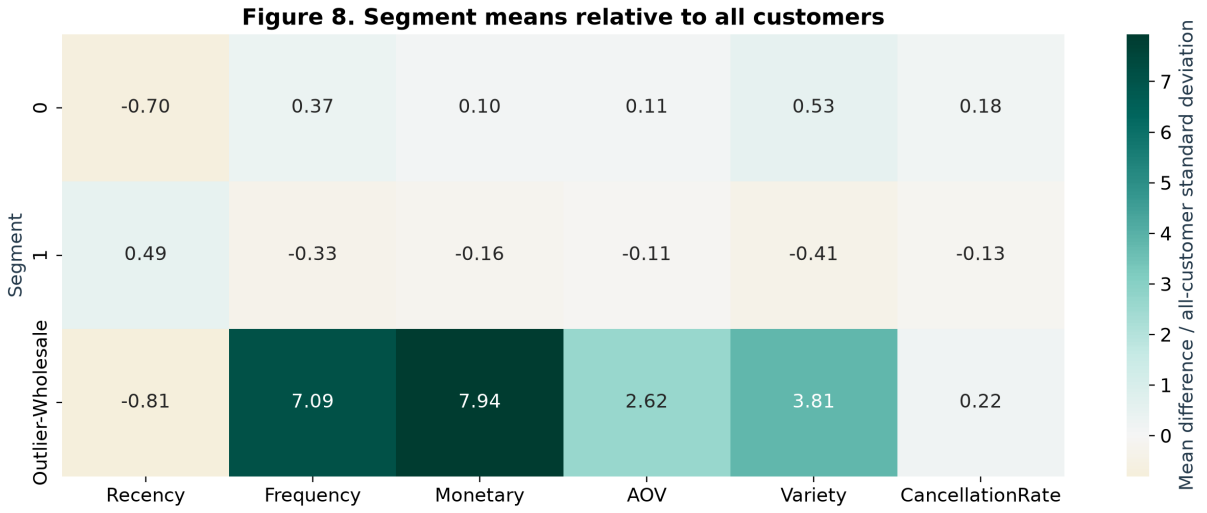
Figure 7. PCA view of fitted customers



PCA displays 94.8% of transformed feature variance. Colours are pipeline predictions. Projection overlap is not a separate clustering evaluation; outliers were excluded from fitting.

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Figure 8. Segment means relative to all customers



Profiles cover 5,839 customers and 3 groups including Outlier-Wholesale. Cells standardise raw segment means using the all-customer mean and standard deviation; the outlier label does not establish a wholesale business model.

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Full model selection results

Feature set	k	Inertia	Silhouette	Davies-Bouldin	Seed ARI
rfm	2	8,375.49	0.438	0.873	0.999
rfm	3	6,189.88	0.346	1.046	0.988
rfm	4	4,734.97	0.368	0.928	0.989
rfm	5	4,009.05	0.334	0.975	0.919
rfm	6	3,449.85	0.335	0.952	0.984
rfm	7	3,090.72	0.297	0.976	0.935
rfm	8	2,801.17	0.297	0.991	0.954
rfm	9	2,571.83	0.295	0.998	0.958
rfm	10	2,387.87	0.283	1.011	0.979
extended	2	11,707.61	0.409	0.925	0.999
extended	3	8,870.54	0.314	1.104	0.993
extended	4	7,391.20	0.312	1.102	0.990
extended	5	6,432.47	0.266	1.124	0.986
extended	6	5,772.39	0.260	1.137	0.835
extended	7	5,189.49	0.257	1.114	0.961
extended	8	4,859.43	0.245	1.165	0.822
extended	9	4,567.56	0.251	1.150	0.865
extended	10	4,279.69	0.242	1.133	0.921

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